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| **20AM6L03 - R-PROGRAMMING LAB** | | | | |

**Prerequisites:**

Any Programming Language.

**COURSE OUTCOMES :**

At the end of the Course, the Student will be able to:

**CO1:** Setup R Programming Environment.

**CO2:** Understand and use R – Data types.

**CO3:** Understand and use R – Data Structures.

**CO4:** Develop programming logic using R – Packages.

**CO5:** Analyze data sets using R – programming capabilities

**LIST OF EXPERIMENTS:**

1. Download and install R-Programming environment and install basic packages using install.packages() command in R.
2. Learn all the basics of R-Programming (Data types, Variables, Operators etc,.)
3. Write a program to find list of even numbers from 1 to n using R-Loops.
4. Create a function to print squares of numbers in sequence.
5. Write a program to join columns and rows in a data frame using cbind() and rbind() in R.
6. Implement different String Manipulation functions in R.
7. Implement different data structures in R (Vectors, Lists, Data Frames)
8. Write a program to read a csv file and analyze the data in the file in R.
9. Create pie chart and bar chart using R.
10. Create a data set and do statistical analysis on the data using R.

**Text Book:**

1. Norman Matloff, The Art of R Programming, UC Davis 2009.

**Web Reference:**

1. https://www.r-project.org/